Marking Scheme
Strictly Confidential
(For Internal and Restricted use only)
Secondary School Supplementary Examination, July- 2023
SUBJECT NAME: SCIENCE SUBJECT CODE : 086 PAPER CODE : 31/C/1

General Instructions: -

1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
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3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.**
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9	If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note

	“Extra Question”.
10	No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
11	A full scale of marks _____80_____(example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer deserves it.
12	Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past:- Giving more marks for an answer than assigned to it. ● Wrong totaling of marks awarded on an answer. ● Wrong transfer of marks from the inside pages of the answer book to the title page. Wrong question wise totaling on the title page. ● Leaving answer or part thereof unassessed in an answer book. ● Wrong totaling of marks of the two columns on the title page. ● Wrong grand total. ● Marks in words and figures not tallying/not same. ● Wrong transfer of marks from the answer book to online award list. ● Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.) ● Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
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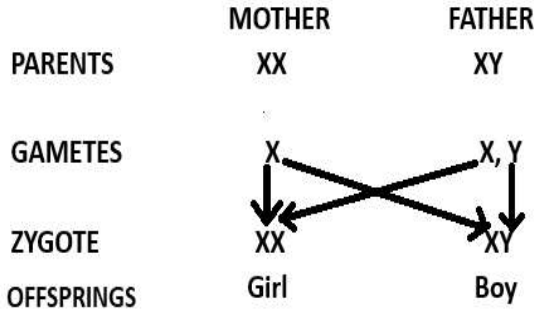
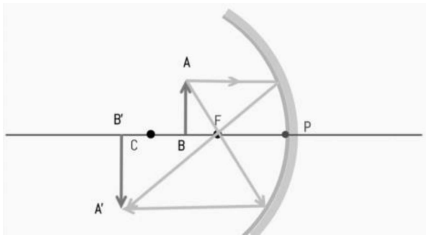
MARKING SCHEME
Secondary School Supplementary Examination, July - 2023
SCIENCE (Subject Code–086)
[Paper Code: 31/C/1]

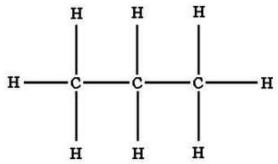
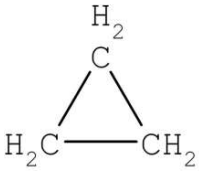
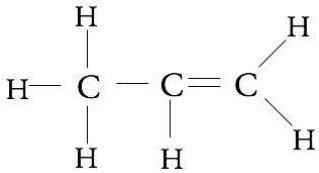
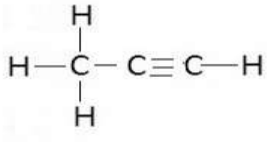
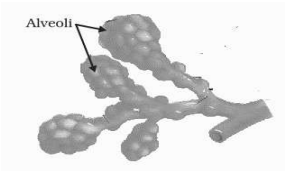
Maximum Marks : 80

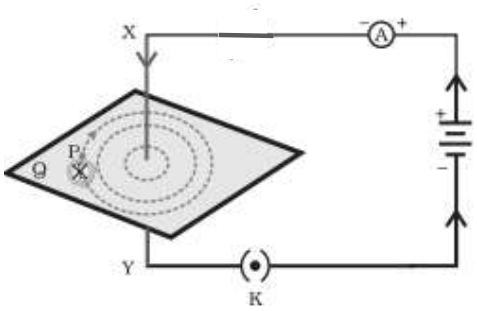
Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
	SECTION—A		
1	(d)	1	1
2	(a) / (c)	1	1
3	(c)	1	1
4	(b)	1	1
5	(c)	1	1
6	(b)	1	1
7	(c)	1	1
8	(d)	1	1
9	(c)	1	1
10	(d)	1	1
11	(a)	1	1
12	(a)	1	1
13	(d)	1	1
14	(a)	1	1
15	(c)	1	1
16	(a)	1	1
17	(d)	1	1
18	(b)	1	1
19	(c)	1	1
20	(d)	1	1
	SECTION— B		
21	(a) A = Potassium / K **or** Sodium /Na B = Calcium / Ca **or** Magnesium / Mg C = Aluminium / Al **or** Iron / Fe **or** Zinc / Zn D = Lead / Pb **or** Copper / Cu **or** Silver / Ag **or** Gold / Au	½ ×4	

	OR		
	(b) (i) (1) Sodium chloride -yellow	½	
	(2) Calcium chloride – brick red	½	
	(credit full marks for any other colour)		
	(ii) No	½	
	Justification : Because they are ionic or electrovalent compounds which are insoluble in organic solvents	½	2
22	• Prostate gland, and seminal vesicles	½, ½	
	• 1) It provides nutrition to the sperms.	½	
	2) It makes the transport of sperms easier through the fluid medium.	½	2
23			
	Diagram	1	
	Labelling	1	
	(Award 1 mark if explained through words only)		2
24	(a) $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	½	
	$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$		
	$= \frac{1}{-15} - \frac{1}{-20} = \frac{-1}{60}$		
	v = - 60 cm	½	
	m = $\frac{-v}{u}$	½	
	= - $\frac{-60}{-20}$ = -3	½	
	OR		
	(b) 		
	Diagram	1½	2
	Direction of the rays	½	

25	Short circuiting / fuse will blow or melt / break in circuit. (any one) Justification: $I = \frac{P}{V} = \frac{3000\ W}{220\ V} = 13.6\ A$ As the current rating is only 10 A and the oven draws a current of 13.6 A, which is more than the current rating. Hence the oven will stop working.	1 1	 2
26	(a) Cloth bag is biodegradable / eco-friendly. (b) i. segregation of biodegradable and non-biodegradable waste at source. ii. By composting iii. Recycling of solid wastes (any two) (any other suitable way)	1 ½+½	 2
	SECTION—C		
27	• The process in which plants/autotrophs take in substances from outside and convert them into stored form of energy. / The process in which CO_2 and water is converted into carbohydrates in the presence of sunlight and chlorophyll. / $6CO_2 + 12H_2O \xrightarrow[\text{Sunlight}]{\text{Chlorophyll}} \underset{\text{(Glucose)}}{C_6H_{12}O_6} + 6O_2 + 6H_2O$ • Chloroplast • leaves / green part of the plant. • water • stored as starch	 1 ½ ½ ½ ½	 3
28	(a) (i) 'X' is Ethanol / C_2H_5OH (ii) • $2C_2H_5OH + 2Na \longrightarrow 2C_2H_5O^-Na^+ + H_2$ $C_2H_5OH \xrightarrow[443K]{H_2SO_4(conc)} CH_2{=}CH_2 + H_2O$ (Do not deduct marks if equations are not balanced) • Sulphuric acid acts as a dehydrating agent. **OR** (b) (i) due to the formation of strong covalent bonds. (ii)	 1 1 ½ ½ 1	

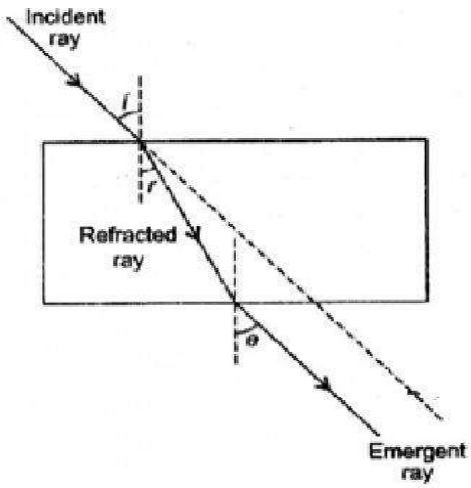
Saturated compounds	**Unsaturated compounds**
• Compounds containing	• Compounds containing carbon-

	carbon- carbon single bonds. / carbon double or triple bonds. (Any other difference) • Structure of saturated compound  /  • Structure of unsaturated compound  /  (any one in each)	1 ½ ½	3
29	(a) Within the lungs when the air passage divides into smaller and smaller tubes which finally terminate into balloon- like structures, these are called alveoli. /  Function – It provides a surface where the exchange of gases takes place. (b) It is the volume of air left in the lungs after exhalation.	1 1 1	3
30	• Exchange of gases through stomata. • By the process of respiration. • The guard cells absorb water and swell causing the stomatal pore to open. • The guard cells lose water and shrink and hence the pore closes.	½ ½ 1 1	3
31	• It is the phenomenon of scattering of light by the colloidal particles due to which the path of the light becomes visible. • Examples: 1. When sunlight passes through a canopy of dense forest. 2. A fine beam of sunlight enters a smoke/ dust filled dark room through a small hole. • Explanation: When a beam of light passes through a heterogenous mixture of minute particles (dust, smoke etc) it is reflected diffusely by the particles and gets scattered. This scattering of light makes the beam of light visible.	1 ½ ½ 1	3

32	(a) (i) Solenoid : A coil of many turns of insulated copper wire wrapped closely in the shape of cylinder. Circular coil : Straight wire bent in the form of circular loop with many turns. (ii) By taking a non-conducting cylindrical tube and winding a long, insulated copper wire tightly over it in the shape of a spring such that the turns are closely placed and lie side by side. (iii) To magnetise a piece of magnetic material like soft iron / To make an electromagnet (any one) **OR** (b) •  Diagram Labelling of magnetic field (Please check the direction of magnetic field lines corresponding to the direction of current in the conductor.) • Right hand Thumb Rule : Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field.	1 1 1 1 1 1	3
33	• Trophic level – Various steps or levels in a food chain • (i) secondary consumer – third trophic level. (ii) tertiary consumer – fourth trophic level. • 1. Death of all organisms at one trophic level will lead to ecological imbalance / disrupt the food chain. 2. Organisms of the next level will starve to death / enter other food chain / organisms of the previous level will multiply profusely / (any other)	1 ½ ½ ½ ½	3
	SECTION—D		
34	(a) (i) (2) Magnesium hydroxide (ii) (2) $Ca(OH)_2$ and (4) NaOH	1 ½+ ½	

	(iii) • NH_3 / Ammonia • NH_4OH / Ammonium hydroxide (iv) To neutralize the effect of acid in the bee sting. (v) (1) Oxalic acid (2) Tartaric acid **OR** (b)(i) It is the fixed number of water molecules present in one formula unit of salt. (ii) Hydrated copper sulphate / Copper sulphate penta hydrate. $CuSO_4.5H_2O$ (iii) • $CaOCl_2$ • Chemical equation $Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ • Uses – 1. For bleaching cotton and linen in textile industry. 2. As an oxidising agent in a chemical industry. 3. For disinfecting water. (or any other)	½ ½ 1 ½+ ½ 1 ½ ½ ½ 1 ½ ×3	5
35	(a) (i) (1) Ovary (2) Oviduct / Fallopian tube (3) Lining of the uterus (ii) (1) zygote is formed (2) when egg is not fertilised, egg lives for about one day, the lining of the uterus slowly breaks down and comes out through vagina along with blood and mucous. **OR** (b) (i) (1) Unisexual flower – contains either stamens or pistil. eg: Papaya/ Water melon (any other) (2) Bisexual flower – contains both stamens and pistil eg: Hibiscus / mustard (any other) (ii) A – Pollen Grain B – Stigma C – Pollen tube D – Female germ-cell / egg cell (iii) Transfer of pollen is required for fusion of gametes. / Pollen needs to be transferred from the stamen to the stigma as it brings male germ-cell (Pollen) + female germ-cell (egg) together for fusion.	1 1 1 1 1 ½ ½ ½ ½ ½ ×4 1	5

36	(a) • kilowatt hour or kWh • 1 kWh = 1000W × 3600 seconds = 3.6×10^6 watt seconds 1 kWh = 3.6×10^6 Joule (b) P = 8 W ; V = 220 V ; Current rating = 1.0 A Current through each lamp = $\frac{8\,W}{220\,V}$ Number of lamp = $\frac{current\ rating}{current\ through\ each\ lamp}$ $= \frac{1.0A}{\frac{8W}{220\,V}} = \frac{220}{8} = 27.75$ So, 27 lamps can safely be used in the circuit.	1 ½ 1 1 ½ 1	5
	SECTION - E		
37	(a) • Middle • Sulphides /Carbonates/Oxide (any one) (b) **Roasting** / **Calcination** Ore is heated in excess of air. / Ore is heated in the absence or limited supply of air. This is used for sulphide ores. / This is used for carbonate ores. (Any one difference) (c) • Galvanization – coating of iron object with a thin layer of zinc. • Alloying – A mixture of two or more metals or a metal and a non-metal. (or any other) **OR** (c) The reaction between aluminium and iron oxide is highly exothermic which forms molten iron which is used in welding cracked machine parts. $Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s) + Heat$	½ ½ 1 1 1 1 1	4
38	(a) • Bony box / Cranium, fluid filled balloon (b) • i. Sensory neuron – pass information from receptors to spinal cord. ii. Motor neuron – transmit information from spinal cord to effector organ / muscle. (c) (i) Cerebellum / Hind Brain – Voluntary Action (ii) Medulla / Hind Brain - Involuntary Action	½, ½ ½ ½ ½+ ½ ½+ ½	

	OR (c) Through Peripheral nervous system • Cranial nerves, and Spinal nerves	1 ½, ½	4
39	(a) The refractive index of a medium with respect to air or vacuum / Absolute refractive index of a medium = $\frac{\text{speed of light in air(vacuum)}}{\text{speed of light in medium}}$ (b) (i) speed of light is more in water (ii) bends away from normal (c) Absolute refractive index of a medium = $\frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$ $\frac{3}{2} = \frac{\text{speed of light in vacuum}}{2 \times 10^8}$ Speed of light in vacuum = 3×10^8 m/s Speed of light in water = $\frac{3}{4} \times 3 \times 10^8$ m/s = $\frac{9}{4} \times 10^8$ m/s **OR** (c)  Diagram Labelling	1 ½ ½ ½ ½ 1 1 1	4

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SUBJECT NAME: SCIENCE SUBJECT CODE : 086 PAPER CODE : 31/C/2

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MARKING SCHEME
SECONDARY SCHOOL SUPPLEMENTARY EXAMINATION - JULY, 2023
SUBJECT : SCIENCE CODE–086
[PAPER CODE: 31/C/2]

Maximum Marks:80

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
	SECTION—A		
1	(b)	1	1
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3	(d)	1	1
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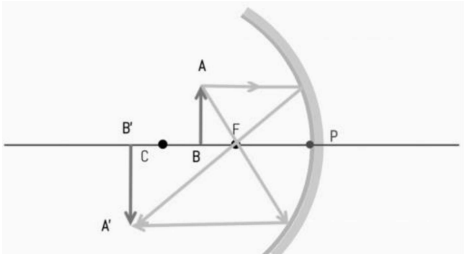
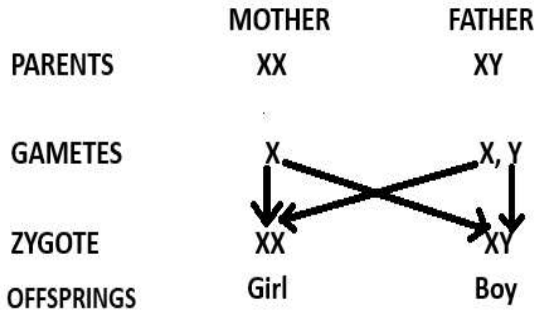
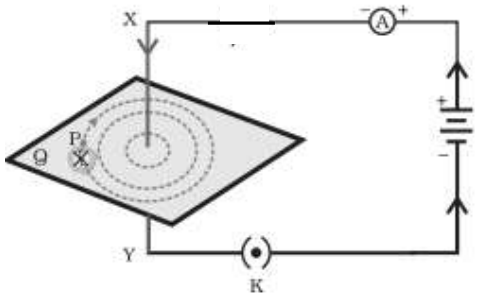
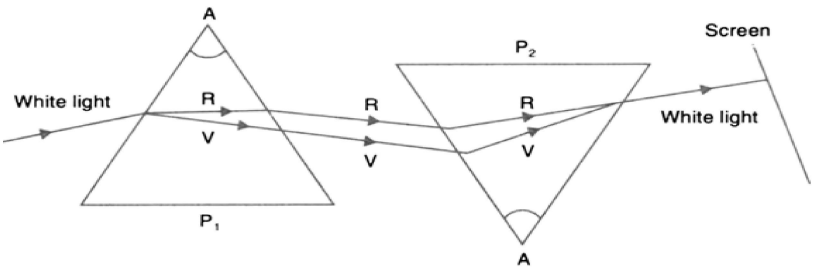
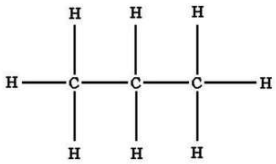
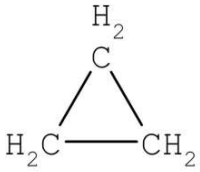
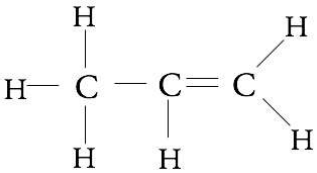
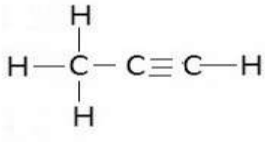
5	(d)	1	1
6	(c)	1	1
7	(a)	1	1
8	(d)	1	1
9	(a)	1	1
10	(c)	1	1
11	(a)	1	1
12	(c)	1	1
13	(d)	1	1
14	(c)	1	1
15	(d)	1	1
16	(a)	1	1
17	(a)	1	1
18	(a)	1	1
19	(d)	1	1
20	(c)	1	1
	SECTION—B		
21	(a) $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$	½	

	$= \frac{1}{-15} - \frac{1}{-20} \quad = \frac{-1}{60}$ v = - 60 cm m = $\frac{-v}{u}$ $= - \frac{-60}{-20} = -3$ **OR** (b)  Diagram Direction of the rays	½ ½ ½ 1½ ½	2
22	• Leaves of Bryophyllum have buds in their notches along the leaf margin whereas leaves of lemon plant do not have such buds. • Method- vegetative propagation.	1 1	2
23	(a) Cloth bag is biodegradable / eco-friendly. (b) i. Segregation of biodegradable and non-biodegradable waste at source. ii. By composting iii. Recycling of solid wastes (any two) (any other suitable way)	1 ½+½	2
24	(a) A = Potassium / K **or** Sodium /Na B = Calcium / Ca **or** Magnesium / Mg C = Aluminium / Al **or** Iron / Fe **or** Zinc / Zn D = Lead / Pb **or** Copper / Cu **or** Silver / Ag **or** Gold / Au **OR** (b) (i) (1) Sodium chloride -yellow (2) Calcium chloride – brick red (credit full marks for any other colour) (ii) No Justification : Because they are ionic or electrovalent compounds which are insoluble in organic solvents.	½ ×4 ½ ½ ½ ½	2
25	• DC always flow in one direction while AC reverses its direction periodically. • AC /Alternating Current	1 ½	

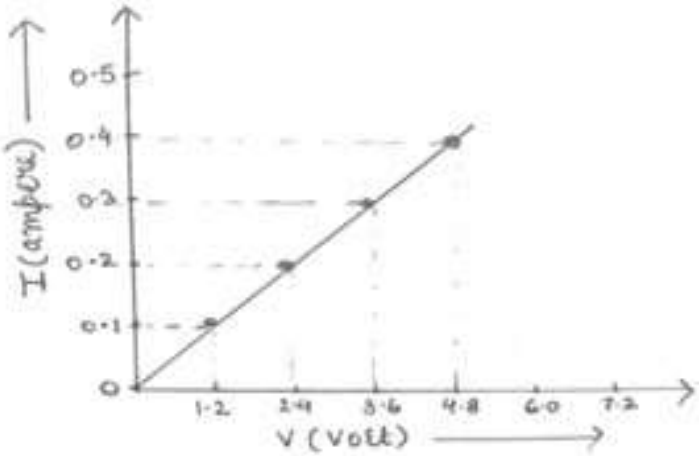
	• 50 Hz	½	2
26	 Diagram Labelling (Award 1 mark if explained through words only)	1 1	2
	SECTION—C		
27	(a) $CaO + H_2O \longrightarrow Ca(OH)_2$ (b) • Combination reaction – A single product is formed from two reactants. • Exothermic reaction – Large amount of heat is evolved.	1 ½ , ½ ½ , ½	3
28	• Exchange of gases through stomata. • By the process of respiration. • The guard cells absorb water and swell causing the stomatal pore to open. • The guard cells lose water and shrink and hence the pore closes.	½ ½ 1 1	3
29	• Through blood as haemoglobin in RBC's present in blood has high affinity for oxygen. • In the dissolved form in the blood.	2 1	3
30	(a) (i) Solenoid : A coil of many turns of insulated copper wire wrapped closely in the shape of cylinder. Circular coil : Straight wire bent in the form of circular loop with many turns. (ii) By taking a non-conducting cylindrical tube and winding a long, insulated copper wire tightly over it in the shape of a spring such that the turns are closely placed and lie side by side. (iii) To magnetise a piece of magnetic material like soft iron / To make an electromagnet (any one) **OR**	1 1 1	

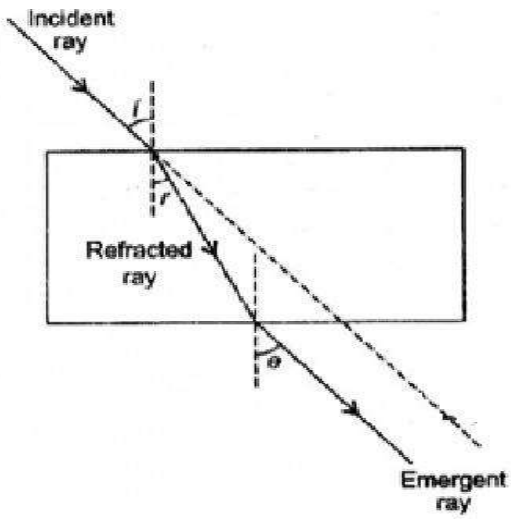
	(b) •  Diagram Labelling of magnetic field (Please check the direction of magnetic field lines corresponding to the direction of current in the conductor.) Right hand Thumb Rule : Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field.	1 1 1	3
31	(a) (i) Violet (ii) Red (b) By placing another identical prism in an inverted position with respect to the first prism.  	½ ½ ½ 1½	3
32	(a) (i) 'X' is Ethanol / C_2H_5OH (ii) • $2C_2H_5OH + 2Na \longrightarrow 2C_2H_5O^-Na^+ + H_2$ $C_2H_5OH \xrightarrow[443K]{H_2SO_4(conc)} CH_2{=}CH_2 + H_2O$ (Do not deduct marks if equations are not balanced) • Sulphuric acid acts as a dehydrating agent. **OR** (b) (i) due to the formation of strong covalent bonds. (ii)	1 1 ½ ½ 1	

Saturated compounds	**Unsaturated compounds**
• Compounds containing	• Compounds containing carbon-

	carbon- carbon single bonds. / carbon double or triple bonds. (Any other difference)	1	
	• Structure of saturated compound  / 	½	
	• Structure of unsaturated compound  /  (any one in each)	½	3
33	(a) • Grass → Grass hopper → Frog → Snake (or any other chain) • Energy at the second trophic level = 7500 J • Justification – Only 10 % of energy is available for transfer to the next level of consumers. / 10% Law	1 ½ ½	
	(b) The energy that is captured by autotrophs does not revert back to the solar input. / The energy which is passed to the herbivores does not come back to autotrophs.	1	3
	SECTION D		
34.	(a) (i) (1) Ovary (2) Oviduct / Fallopian tube (3) Lining of the uterus	1 1 1	
	(ii) (1) Zygote is formed (2) When egg is not fertilised, egg lives for about one day, the lining of the uterus slowly breaks down and comes out through vagina along with blood and mucous.	1 1	
	OR		
	(b) (i) (1) Unisexual flower – contains either stamens or pistil. eg: Papaya/ Water melon (any other) (2) Bisexual flower – contains both stamens and pistil eg: Hibiscus / mustard (any other)	½ ½ ½ ½	
	(ii) A – Pollen Grain B – Stigma		

	C – Pollen tube D – Female germ-cell / egg cell (iii) Transfer of pollen is required for fusion of gametes. / Pollen needs to be transferred from the stamen to the stigma as it brings male germ-cell (Pollen) + female germ-cell (egg) together for fusion.	½ ×4 1	5
35	(a) (i) (2) Magnesium hydroxide (ii) (2) $Ca(OH)_2$ and (4) NaOH (iii) • NH_3 / Ammonia • NH_4OH / Ammonium hydroxide (iv) To neutralize the effect of acid in the bee sting. (v) (1) Oxalic acid (2) Tartaric acid **OR** (b)(i) It is the fixed number of water molecules present in one formula unit of salt. (ii) hydrated copper sulphate / copper sulphate penta hydrate. $CuSO_4.5H_2O$ (iii) • $CaOCl_2$ • Chemical equation $Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ • Uses – 1. For bleaching cotton and linen in textile industry. 2. As an oxidising agent in a chemical industry. 3. For disinfecting water. (or any other)	1 ½+ ½ ½ ½ 1 ½+ ½ 1 ½ ½ ½ 1 ½ ×3	5
36	(a) l = 1 m $A = 2mm^2 = 2 \times (10^{-3}m)^2 = 2 \times 10^{-6} m^2$ $\rho = 2.63 \times 10^{-8}\ \Omega m$ $R = \rho \frac{l}{A}$ $R = (2.63 \times 10^{-8}\ \Omega m) \times \frac{1m}{2 \times 10^{-6}\ m^2}$ $= \frac{2.63}{2} \times 10^{-2}$ $= 1.315 \times 10^{-2} \Omega$ (b)	½ ½ 1 ½	

	 Since I – V graph is a straight line passing through the origin it shows that $I \propto V$ which is in accordance with the Ohm's law.	2 ½	5
	SECTION E		
37	(a) • Middle • Sulphides /Carbonates/Oxide (any one) (b) **Roasting** / **Calcination** Ore is heated in excess of air. / Ore is heated in the absence or limited supply of air. This is used for sulphide ores. / This is used for carbonate ores. (Any one difference) (c) • Galvanization – coating of iron object with a thin layer of zinc. • Alloying – A mixture of two or more metals or a metal and a non-metal. (or any other) **OR** (c) The reaction between aluminium and iron oxide is highly exothermic which forms molten iron which is used in welding cracked machine parts. $Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s) + Heat$	½ ½ 1 1 1 1 1	4
38	(a) • Bony box / Cranium, fluid filled balloon (b) • i. Sensory neuron – pass information from receptors to spinal cord. ii. Motor neuron – transmit information from spinal cord to effector organ / muscle. (c) (i) Cerebellum / Hind Brain – Voluntary Action (ii) Medulla / Hind Brain - Involuntary Action **OR** (c) Through Peripheral nervous system	½, ½ ½ ½ ½+ ½ ½+ ½ 1	

	• Cranial nerves and Spinal nerves	½ , ½	4
39	(a) The refractive index of a medium with respect to air or vacuum / Absolute refractive index of a medium = $\frac{\text{speed of light in air(vacuum)}}{\text{speed of light in medium}}$	1	
	(b) (i) speed of light is more in water	½	
	(ii) bends away from normal	½	
	(c) Absolute refractive index of a medium = $\frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$		
	$\frac{3}{2} = \frac{\text{speed of light in vacuum}}{2 \times 10^8}$	½	
	Speed of light in vacuum = 3×10^8 m/s	½	
	Speed of light in water = $\frac{3}{4} \times 3 \times 10^8$ m/s = $\frac{9}{4} \times 10^8$ m/s	1	
	OR		
	(c)  Diagram Labelling	1 1	4

* **

Marking Scheme
Strictly Confidential
(For Internal and Restricted use only)
Secondary School Supplementary Examination, July- 2023

SUBJECT NAME: SCIENCE **SUBJECT CODE-086** **PAPER CODE 31/C/3**

General Instructions: -

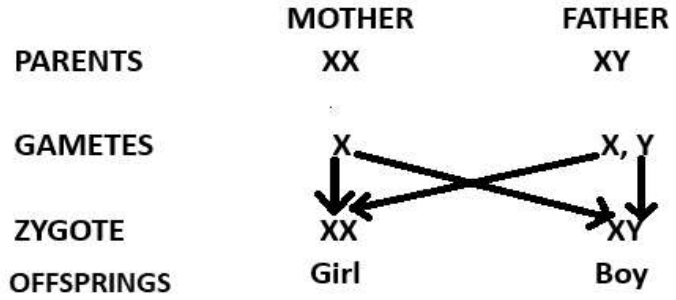
1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2	**"Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its' leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC."**
3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.**
4	The Marking scheme carries only suggested value points for the answers.These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.
5	The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after delibration and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
6	Evaluators will mark(√) wherever answer is correct. For wrong answer CROSS 'X" be marked. Evaluators will not put right (✓)while evaluating which gives an impression that answer is correct and no marks are awarded. **This is most common mistake which evaluators are committing.**
7	If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
8	If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
9	If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note **"Extra Question"**.
10	No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
11	A full scale of marks _____80_____(example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer

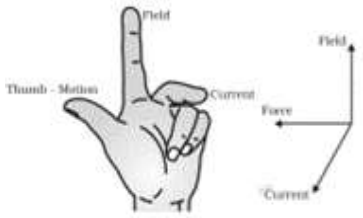
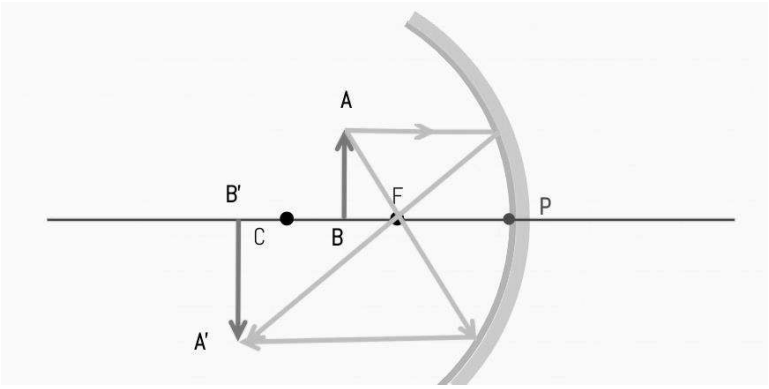
	deserves it.
12	Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past:- Giving more marks for an answer than assigned to it. ● Wrong totaling of marks awarded on an answer. ● Wrong transfer of marks from the inside pages of the answer book to the title page. Wrong question wise totaling on the title page. ● Leaving answer or part thereof unassessed in an answer book. ● ● Wrong totaling of marks of the two columns on the title page. ● Wrong grand total. ● Marks in words and figures not tallying/not same. ● Wrong transfer of marks from the answer book to online award list. ● Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.) ● Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
14	While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
15	Any un assessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
16	The Examiners should acquaint themselves with the guidelines given in the "**Guidelines for spot Evaluation**" before starting the actual evaluation.
17	Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
18	The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.

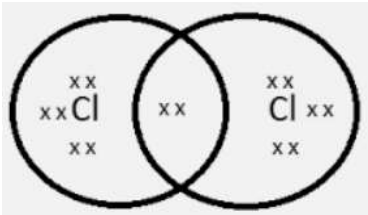
MARKING SCHEME
SECONDARY SCHOOL SUPPLEMENTARY EXAMINATION JULY, 2023
SUBJECT : SCIENCE CODE–086
[PAPER CODE: 31/C/3]

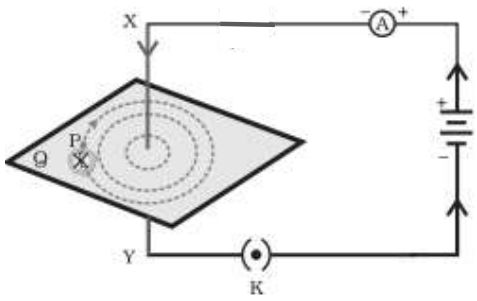
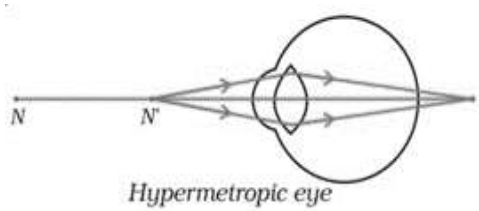
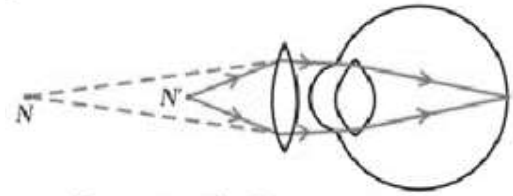
Maximum Marks:80

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
	SECTION—A		
1	(b)	1	1
2	(c)	1	1
3	(b)	1	1
4	(d)	1	1
5	(a) / (c)	1	1
6	(c)	1	1
7	(b)	1	1
8	(c)	1	1
9	(d)	1	1
10	(c)	1	1
11	(c)	1	1
12	(a)	1	1
13	(a)	1	1
14	(b)	1	1
15	(a)	1	1
16	(d)	1	1
17	(d)	1	1
18	(d)	1	1
19	(b)	1	1
20	(b)	1	1

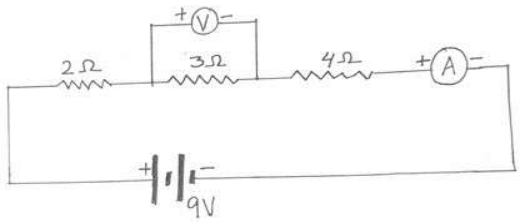
	SECTION—B		
21	 Diagram Labelling (Award 1 mark if explained through words only)	1 1	2
22	• (i) Testis (ii) Ovary • Hormone secreted by : Testis – Testosterone Ovary – Oestrogen/Progesterone	½ ½ ½ ½	2
23	(a) Cloth bag is biodegradable / eco-friendly. (b) i. Segregation of biodegradable and non-biodegradable waste at source. ii. By composting iii. Recycling of solid wastes (any two) (any other suitable way)	1 ½+½	2
24	(a) A = Potassium / K **or** Sodium /Na B = Calcium / Ca **or** Magnesium / Mg C = Aluminium / Al **or** Iron / Fe **or** Zinc / Zn D = Lead / Pb **or** Copper / Cu **or** Silver / Ag **or** Gold / Au **OR** (b) (i) (1) Sodium chloride -yellow (2) Calcium chloride – brick red (credit full marks for any other colour) (ii) No Justification : Because they are ionic or electrovalent compounds which are insoluble in organic solvents.	½ ×4 ½ ½ ½ ½	2
25	(a) • According to Fleming's left hand rule, stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular		

	to each other. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor. /  (b) Into the plane of the paper (downward)	1 1	2
26	(a) $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$ $= \frac{1}{-15} - \frac{1}{-20} \quad = \frac{-1}{60}$ v = - 60 cm m = $\frac{-v}{u}$ $= - \frac{-60}{-20} = -3$ **OR** (b)  Diagram Direction of the rays	½ ½ ½ ½ 1½ ½	2
	SECTION—C		
27	• Exchange of gases through stomata. • By the process of respiration. • The guard cells absorb water and swell causing the stomatal pore to open. • The guard cells lose water and shrink and hence the pore closes.	½ ½ 1 1	` 3

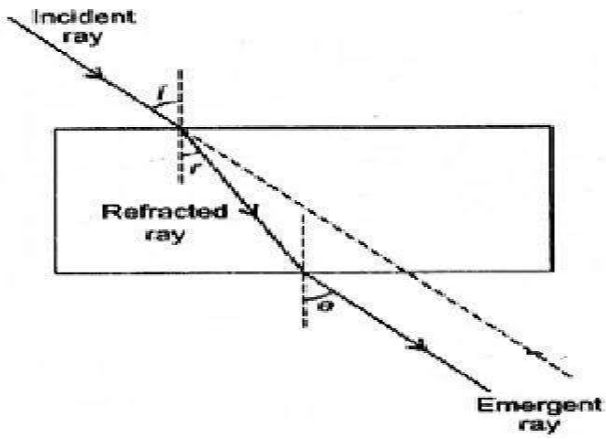
28	(a) (i) The chemical properties are determined solely by the functional group, which remains the same for a particular homologous series. (ii) One carbon, two hydrogen / CH_2 (i) $C_2H_2 + CH_2 \longrightarrow C_3H_4$ (ii) $C_2H_5OH + CH_2 \longrightarrow C_3H_7OH$ (iii) $CH_3CHO + CH_2 \longrightarrow C_2H_5CHO$ (any other) **OR** (b) (i) (1) Small size of carbon atom enables the nucleus to hold on to the shared pair of electrons strongly. (2) Weak inter molecular forces of attraction. (ii) Cl = 17 = 2,8,7 	1 ½ ½ ½ ½ 1 1 1	3
29	(a) • Gastric glands • Pepsin, Mucus, HCl (b) • Emulsification of fats / Bile juice breaks fats into small globules • It makes the food coming from the stomach alkaline for the pancreatic enzymes to act.	½ ½ × 3 ½ ½	3
30	(a) (i) Solenoid : A coil of many turns of insulated copper wire wrapped closely in the shape of cylinder. Circular coil : Straight wire bent in the form of circular loop with many turns. (ii) By taking a non-conducting cylindrical tube and winding a long, insulated copper wire tightly over it in the shape of a spring such that the turns are closely placed and lie side by side. (iii) To magnetise a piece of magnetic material like soft iron / To make an electromagnet **(any one)**	1 1 1	

	OR (b) •  Diagram Labelling of magnetic field (Please check the direction of magnetic field lines corresponding to the direction of current in the conductor.) • Right hand Thumb Rule : Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field.	1 1 1	3
31	• Hypermetropia / far sightedness. • By using a convex lens of appropriate power (i)  Hypermetropic eye (ii)  Correction for Hypermetropic eye	½ ½ 1 1	3
32	• The process in which plants/autotrophs take in substances from outside and convert them into stored form of energy. / The process in which CO_2 and water is converted into carbohydrates in the presence of sunlight and chlorophyll. /		

	$6CO_2 + 12H_2O \xrightarrow[\text{Sunlight}]{\text{Chlorophyll}} C_6H_{12}O_6 + 6O_2 + 6H_2O$ (Glucose) • Chloroplast • leaves / green part of the plant. • water • stored as starch	1 ½ ½ ½ ½	3
33	(a) (i) Forest / Ocean (ii) Aquarium / Crop fields (or any other) (b) • Algae ⟶ Mosquito larvae ⟶ small fish ⟶ Crane (Any other chain) • Producers • As only they can convert sunlight into food energy.	½ ½ 1 ½ ½	3
	SECTION D		
34	(a) (i) (2) Magnesium hydroxide (ii) (2) $Ca(OH)_2$ and (4) NaOH (iii) • NH_3 / Ammonia • NH_4OH / Ammonium hydroxide (iv) To neutralize the effect of acid in the bee sting. (v) (1) Oxalic acid (2) Tartaric acid **OR** (b)(i) It is the fixed number of water molecules present in one formula unit of salt. (ii) hydrated copper sulphate / copper sulphate penta hydrate. $CuSO_4.5H_2O$ (iii) • $CaOCl_2$ • Chemical equation $Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ • Uses – 1. For bleaching cotton and linen in textile industry. 2. As an oxidising agent in a chemical industry. 3. For disinfecting water. (or any other)	1 ½+ ½ ½ ½ 1 ½+ ½ 1 ½ ½ ½ 1 ½ ×3	5

35	(a) (i) (1) Ovary (2) Oviduct / Fallopian tube (3) Lining of the uterus (ii) (1) zygote is formed (2) when egg is not fertilised, egg lives for about one day, the lining of the uterus slowly breaks down and comes out through vagina along with blood and mucous. **OR** (b) (i) (1) Unisexual flower – contains either stamens or pistil. eg: Papaya/ Water melon (any other) (2) Bisexual flower – contains both stamens and pistil eg: Hibiscus / mustard (any other) (ii) A – Pollen Grain B – Stigma C – Pollen tube D – Female germ-cell / egg cell (iii) Transfer of pollen is required for fusion of gametes. / Pollen needs to be transferred from the stamen to the stigma as it brings male germ-cell (Pollen) + female germ-cell (egg) together for fusion.	1 1 1 1 1 ½ ½ ½ ½ ½ ×4 1	5
36	(a) (i) $R_s = R_1 + R_2 + R_3$ (ii) $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$ (b) •  (i) $R_S = 2\,\Omega + 3\,\Omega + 4\,\Omega = 9\,\Omega$ V = 9V $I = \frac{v}{R_S}$	1 1 1	

	$= \frac{9v}{9\Omega} = 1$ A Ammeter reading = 1.0 A (ii) Potential Difference across 3 Ω resistor V = I R = 1.0 A × 3.0 Ω = 3 V	1 1	5
	SECTION E		
37	(a) • Middle • Sulphides /Carbonates/Oxide (any one) (b) **Roasting** / **Calcination** Ore is heated in excess of air. / Ore is heated in the absence or limited supply of air. This is used for sulphide ores. / This is used for carbonate ores. (Any one difference) (c) • Galvanization – coating of iron object with a thin layer of zinc. • Alloying – A mixture of two or more metals or a metal and a non-metal. (or any other) OR (c) The reaction between aluminium and iron oxide is highly exothermic which forms molten iron which is used in welding cracked machine parts. $Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s) + Heat$	½ ½ 1 1 1 1 1	4
38	(a) • Bony box / Cranium, fluid filled balloon (b) • i. Sensory neuron – pass information from receptors to spinal cord. ii. Motor neuron – transmit information from spinal cord to effector organ / muscle. (c) (i) Cerebellum / Hind Brain – Voluntary Action (ii) Medulla / Hind Brain - Involuntary Action **OR** (c) Through Peripheral nervous system • Cranial nerves and Spinal nerves	½, ½ ½ ½ ½+ ½ ½+ ½ 1 ½, ½	4

39	(a) The refractive index of a medium with respect to air or vacuum / Absolute refractive index of a medium = $\frac{\text{speed of light in air(vacuum)}}{\text{speed of light in medium}}$	1	
	(b) (i) speed of light is more in water	½	
	(ii) bends away from normal	½	
	(c) Absolute refractive index of a medium = $\frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$		
	$\frac{3}{2} = \frac{\text{speed of light in vacuum}}{2 \times 10^8}$	½	
	Speed of light in vacuum = 3×10^8 m/s	½	
	Speed of light in water = $\frac{3}{4} \times 3 \times 10^8$ m/s $= \frac{9}{4} \times 10^8$ m/s	1	
	OR		
	(c)  Diagram Labelling	1 1	4

* **

Marking Scheme
Strictly Confidential
(For Internal and Restricted use only)
Secondary School Supplementary Examination, July- 2023
SUBJECT NAME: SCIENCE SUBJECT CODE- 086 PAPER CODE 31(B)

General Instructions: -

1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2	**"Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its' leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC."**
3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.**
4	The Marking scheme carries only suggested value points for the answers.These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.
5	The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after delibration and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
6	Evaluators will mark(√) wherever answer is correct. For wrong answer CROSS 'X" be marked. Evaluators will not put right (✓)while evaluating which gives an impression that answer is correct and no marks are awarded. **This is most common mistake which evaluators are committing.**
7	If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
8	If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
9	If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note **"Extra Question"**.
10	No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
11	A full scale of marks _____80_____(example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer deserves it.
12	Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past:- Giving more marks for an answer than assigned to it. ● Wrong totaling of marks awarded on an answer. ● Wrong transfer of marks from the inside pages of the answer book to the title page.

	Wrong question wise totaling on the title page. ● Leaving answer or part thereof unassessed in an answer book. ● ● Wrong totaling of marks of the two columns on the title page. ● Wrong grand total. ● Marks in words and figures not tallying/not same. ● Wrong transfer of marks from the answer book to online award list. ● Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.) ● Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
14	While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
15	Any un assessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
16	The Examiners should acquaint themselves with the guidelines given in the "**Guidelines for spot Evaluation**" before starting the actual evaluation.
17	Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
18	The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.

MARKING SCHEME
Secondary School Supplementary Examination, 2023
SCIENCE (Subject Code–086)
[Paper Code: 31/B]

Maximum Marks : 80

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
	SECTION—A		
1	(d)	1	1
2	(c)	1	1
3	(c)	1	1
4	(a)	1	1
5	(b)	1	1
6	(c)	1	1
7	(a)	1	1
8	(b)	1	1
9	(d)	1	1
10	(d)	1	1
11	(c)	1	1
12	(b)	1	1
13	(a)	1	1
14	(a)	1	1
15	(d)	1	1
16	(d)	1	1
17	(b)	1	1
18	(a)	1	1
19	(c)	1	1
20	(a)	1	1
	SECTION— B		
21	(a) (i) When zinc is added to the aqueous solution of copper sulphate, zinc sulphate is formed and blue colour of the solution disappears / $Zn + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu$ (any other example)	1	

	(ii) When aqueous solution of sodium sulphate and barium chloride are mixed together an insoluble precipitate of Barium sulphate is formed / $Na_2SO_4(aq) + BaCl_2(aq) \longrightarrow 2NaCl(aq) + BaSO_4(s)$ (any other example) **OR** (b) (i) Zinc displaces iron from ferrous sulphate solution to form zinc sulphate which is colourless (ii) $FeSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Fe(s)$ Pale green colourless	1 1 1	2
22	• Small Intestine • (i) Glucose / sugar (ii) fatty acids and glycerol (iii) amino acids	½ 3 x ½	2
23	• Auxin (marks can be awarded for Cytokinin / Gibberellin) • Tendrils come in contact with support • Auxin diffuses to the other side. • Auxin promotes cell growth in the part away from the support.	½ ½ ½ ½	2
24	• Pancreas performs dual functions (i) secretes hormones – insulin to maintain the level of sugar in blood. (ii) secretes digestive enzymes (trypsin / lipase / amylase) which help in the digestion of food.	1 1	2
25	(a) • Myopia / Near sightedness Causes: i. Elongation of eyeball ii. Excessive curvature of eye lens • Modification – use of concave lens / diverging lens **OR** b. • Hypermetropia / Far sightedness Causes: i. Focal length of the eye lens increases ii. Eyeball becomes too small • convex lens / converging lens	½ ½ ½ ½ ½ ½ ½ ½	2
26	• DDT, Pesticides, Plastics, Glass (any other) (any two) • Since Hawks are at the top of the food chain, thus the concentration of non-biodegradable substance is maximum due to biological magnification.	½ + ½ 1	2
	SECTION—C		
27	(a) (i) Combination reaction (ii) Decomposition reaction (b) (i) Carbon + Oxygen → Carbon dioxide	½ + ½	

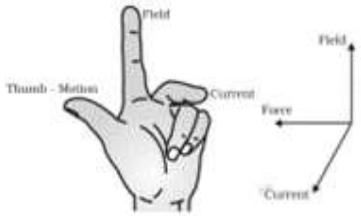
	or C (s) + O_2 (g) $\rightarrow$ CO_2 (g) (or any other example) (ii) Calcium carbonate $\xrightarrow{heat}$ Calcium oxide + Carbon dioxide **or** $CaCO_3$ (s) $\xrightarrow{heat}$ CaO (s) + CO_2 (g) (or any other example)	1 1	3
28	(a) pH paper becomes red in colour - conc. H_2SO_4 pH paper becomes blue in colour - conc. NaOH solution pH paper becomes green in colour - conc. Na_2SO_4 solution. (b) Conc. solution of NaOH and conc. H_2SO_4 (c) H_2SO_4 + 2NaOH $\rightarrow$ Na_2SO_4 + $2H_2O$ (Do not deduct marks if equation is not balanced)	½ ½ ½ 1 ½	3
29	(a) Three events during photosynthesis (i) Absorption of light energy by chlorophyll (ii) Conversion of light energy into chemical energy and splitting of water molecules into hydrogen and oxygen. (iii) Reduction of CO_2 into carbohydrates. **OR** (b) • The colourless fluid present in the intercellular spaces containing some amount of plasma, proteins and blood cells. (any one difference) • Two functions of lymph :- (i) Carries digested and absorbed fat from the intestine. (ii) Drains excess fluid from extra cellular space back into the fluid.	1 1 1 1 1 ½ ½	3

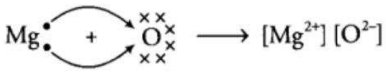
Lymph	Blood Plasma
Colourless Less proteins	Coloured More proteins

30	(a) magnification $= \frac{v}{u}$ $= \frac{+36}{-18}$ m $= -2$ (b) Lens formula : $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ $f = \frac{-18\text{ cm} \times 36\text{ cm}}{-18\text{ cm} - 36\text{ cm}} = \frac{18 \times 36\text{ cm}}{-18\text{ cm} - 36\text{ cm}} = \frac{18 \times 36}{54}$ cm f = 12 cm	½ 1 ½ 1	3
31	(a) (i) Centre of the reflecting surface of a spherical mirror (ii) Centre of the sphere of which mirror was a part		

	(iii) Straight line passing through pole and centre of curvature of spherical mirror (iv) Point on principal axis where the reflected rays meet / appear to meet. (b) Focal length = $\frac{-60\text{ cm}}{2} = -30\text{ cm}$ Because the object is placed at the centre of curvature.	4 x ½ ½ ½	3
32	(a) • The potential difference V, across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it, provided its temperature remains constant / V α I Factors: • Length of conductor. • Area of cross – section of conductor • If length is doubled, cross-sectional area is halved. $R \alpha \frac{l}{A}$ So, resistance becomes 4 times Resistance of new wire = 4R **OR** (b) (i) Formula for Equivalent resistance in series : $R_s = R_1 + R_2 + R_3$ Equivalent resistance in parallel : $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ (ii) (1) Each device can draw current as per need (2) Separate 'ON/OFF' switch for every device can be put (3) Total resistance of the combination decreases hence larger current can be drawn from the source. (4) If one device fails, other devices keep on working (any two)	1 ½ ½ 1 1 1 2 x ½	3
33	• All the organisms (biotic components) in an area interacting together with the non living (abiotic) components of the environment forms an ecosystem. • Two artificial ecosystems – Garden / pond / aquarium (any two) (or any other) • Natural ecosystems are self-sustaining and have decomposers as their cleaning agents whereas the man-made systems do not have them.	1 ½ + ½ 1	3
	SECTION—D		
34	(a) • Methane – CH_4 , Ethane - C_2H_6 • C_nH_{2n+2}	½ + ½ ½	

	• 14 u (b) • Ethanol, C_2H_5OH / CH_3CH_2OH • $C_2H_5OH \xrightarrow[443K]{H_2SO_4(conc)} C_2H_4 + H_2O$ • Ethene • Conc. H_2SO_4 – Dehydrating agent **OR** (a) i. $C_2H_5OH \xrightarrow[\text{or Alk } KMnO_4+heat]{\text{Acidified } K_2Cr_2O_7+heat} CH_3COOH$ ii. $C_2H_4 + H_2 \xrightarrow{\text{Ni catalyst}} C_2H_6$ (b) i. When ethanol is burnt in oxygen then water , carbon dioxide and heat is produced. $C_2H_5OH + 3O_2 \longrightarrow 3H_2O + 2CO_2 + heat$ (any other example) ii. When chlorine reacts with methane in the presence of sunlight, chloromethane is formed. $CH_4 + Cl_2 \xrightarrow{light} CH_3Cl + HCl$ (any other example) iii. Reaction in which an ester reacts with sodium hydroxide to form soap. $CH_3COC_2H_5 + NaOH \rightarrow CH_3COONa + C_2H_5OH$ (any other example) (Do not deduct marks for unbalanced equation.)	½ ½ + ½ 1 ½ ½ 1 1 1 1 1	5
35	(a) • The ability of an organism to give rise to new organisms from their cut or broken body parts /pieces. • Specialised cells . • Planaria / Hydra (any one example) (b) (i) To transport sperms to urethra (ii) To produce sperms / secretion of Testosterone. (iii) To make transport of sperms easier / provide nutrition to the sperms. **OR** (a) • Pollination: Transfer of pollen from anther to the stigma. • Types: Self -Pollination and Cross -Pollination • Self -Pollination: Transfer of pollen occurs in same flower. Cross -Pollination: Transfer of pollen from one flower to another flower. (any other difference)	1 ½ ½ 1 1 1 1 ½ + ½ 1	

	(b) The pollen produces a tube which travels through the style and reaches the ovary. (c) (i) Ovary develops into a fruit. (ii) Ovule develops into a seed.	1 ½ ½	5
36	(a) • Solenoid –A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder. • Inside the solenoid. • Parallel straight magnetic field lines. (b) • Fleming's Left Hand Rule: stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular to each other. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor. /  • Upwards.	1 1 1 1 1	5
	SECTION - E		
37	(a) The refractive index of a medium with respect to air or vacuum / Absolute refractive index of a medium = $\frac{\text{speed of light in air(vacuum)}}{\text{speed of light in medium}}$ (b) (i) water (ii) carbon disuphide (c) (i) $n_g = \frac{3}{2}$; $n_w = \frac{4}{3}$; $v_w = 2.25 \times 10^8$ m/s ; c = ?; v_g = ? (I) $n_w = \frac{c}{v_w}$ $\frac{4}{3} = \frac{c}{2.25 \times 10^8 \text{ m/s}}$ $c = 3 \times 10^8$ m/s (II) $v_g = \frac{c}{n_g}$ $= \frac{3 \times 10^8 \, m/s}{3/2} = 2 \times 10^8 \, m/s$ **OR** (ii) Refractive index of water w.r.t. glass = $\frac{\text{speed of light in glass}}{\text{speed of light in water}}$	1 ½ + ½ ½ ½ ½ ½	

	Absolute refractive index of a medium $= \frac{\text{speed of light in vacuum}}{\text{speed of light in the medium}}$ $\therefore$ speed of light in glass $= \frac{c}{\text{Absloute refractive index of glass}} = \frac{c}{3/2} = \frac{2c}{3}$ speed of light in water $= \frac{c}{\text{Absloute refractive index of water}} = \frac{c}{4/3} = \frac{3c}{4}$ $\therefore$ Refractive index of water w.r.t. glass $= \frac{2c/3}{3c/4} = \frac{8}{9}$	½ ½ ½ ½	4
38	(a) • Electronic configuration of sodium Na (11) = 2, 8, 1 • one electron (b) • Electronic configuration of chlorine Cl (17) =2, 8, 7 • Argon (c) (i) $\underset{2,8,2}{Mg} \rightarrow \underset{\text{Magnesium ion}}{Mg^{2+}} + 2e$ $\underset{2,6}{O} + 2e \rightarrow \underset{\text{Oxygen ion}}{O^{2-}}$  **OR** (ii) (I) Cation : Na^+ [Sodium ion] Anion : Cl^- [Chloride ion] (II) Cation : K^+ [Potassium ion] Anion : NO_3^- [Nitrate ion]	½ ½ ½ ½ ½ ½ 1 ½ ½ ½ ½	4
39	(a) • Capillaries • To bring blood in contact with cells / Exchange of material between the blood and surrounding tissues. (b) To prevent blood loss by clotting of the blood at point of injury. (c) (i) (any other difference) **OR** (ii) In one complete cycle of blood circulation, the blood passes through the heart twice. The deoxygenated blood enters right atrium of the heart from the body and oxygenated blood enters left atrium of the heart from the lungs to complete one cycle.	½ ½ 1 1+1 2	4

Artery	Vein
(i) Thick walled	Thin walled
(ii) Valves absent	Valves present

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